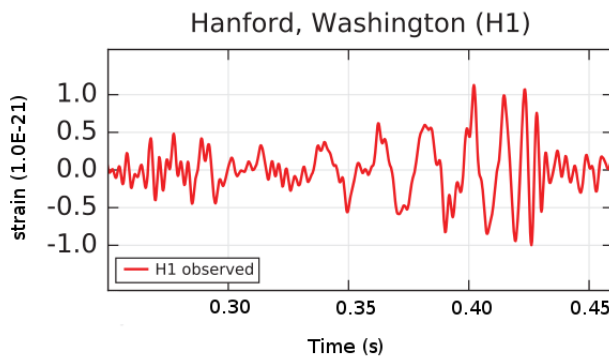


Due Sunday October 18th

1. One of the most significant findings in observational astrophysics this century was the Nobel Prize-winning discovery of gravitational waves in 2015¹. The signal that provided evidence for gravitational waves is shown in the figure below.



peak	time (sec)	δt (sec)
1	0.31172	0.0015
2	0.33927	0.0015
3	0.36113	0.0005
4	0.38370	0.0005
5	0.40201	0.0005
6	0.41481	0.0005
7	0.42316	0.0005

On the right is a table with data taken from this figure. The three columns are (1) the number of the “peak” in the strain, (2) the time at which the peak occurred, and (3) the uncertainty in the timing of the peak. You should assume that these errors are normally distributed. (Disclaimer: points were pulled off the plot and errors estimated by JB, and are not endorsed by LIGO.)

We would like to compare two models for this waveform. Suppose that within General Relativity (GR) the timing of the peaks is modeled by a third-order polynomial, *i.e.*

$$t = a_0 + a_1n + a_2n^2 + a_3n^3$$

where t is the time and n is the number of the peak. In a proposed extension of GR – let's call it GR++ – a fourth-order polynomial is used to model the peaks, *i.e.*

$$t = a_0 + a_1n + a_2n^2 + a_3n^3 + a_4n^4$$

Your task is to find the best-fit parameters a_i for both GR and GR++ models, and use what you know about goodness-of-fit to choose one of the models. Write a Python script `hanford_test.py` that outputs the following:

¹B.P. Abbott *et al.*, Phys. Rev. Lett. **116** 061102 (2016)

```
best-fit value of a_0 from GR:   XXX+/-XXX
best-fit value of a_0 from GR++: XXX+/-XXX
chi2 from GR:   XXX
chi2 from GR++: XXX
Goodness-of-fit Q from GR:   XXX
Goodness-of-fit Q from GR++: XXX
best-pick model: GR/GR++
```

Thus, you are to give the best-fit value of the constant term in both models, χ^2 values, and goodness-of-fit values. On the basis of the goodness-of-fit choose one model or the other as your best-pick, namely the one that you think best describes the data. In coming up with the best-pick model, apply a sharp “Occam’s razor”: Cut out all but the simplest of models, unless otherwise warranted by the data.

I recommend that as a check you plot the points versus the fit results to make sure everything is working. But you don’t need to generate any plots in the final code you submit.

2. This assignment is to be submitted in the form of a Jupyter Notebook **binomial.ipynb**. Please be sure that the parts below are labeled so that the grader can follow your work. Consider a binomial distribution, which describes the probability of k successes in n trials as follows:

$$f(k|M(p)) = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

where the parameter p is the probability of success per trial. In a coin toss, p may be referred to as the “fairness” of the coin, with $p = 0.5$ designated a fair coin and p approaching 0 or 1 designated an unfair coin.

Suppose we flip a coin n times and observe heads k times. What is our best guess of the fairness of the coin and its uncertainty?

- (a) Calculate analytically Fisher information for n binomial RVs. Show all work in a “markdown” cell in your notebook. (Please ask the instructor if you have any questions about how to do this.)
- (b) In a different markdown cell, calculate the MLE $\hat{p}$ for the bias parameter p .
- (c) Use the Cramer-Rao inequality to set a lower bound on the variance and standard deviation of $\hat{p}$.
- (d) Now, we are ready to do a numerical study of our coin flip problem. Write Python code to perform N trials, where in each trial a coin with ground truth fairness p_0 is flipped n times, and the number of heads k is recorded. Do this for different values of the ground truth p_0 .
- (e) for $n = 10$ and $N = 1,000$ (or more), plot mean $\hat{p}$ versus p_0 between 0.0 and 1.0. Comment: Is there any evidence for bias in the MLE?
- (f) for $n = 10$ and $N = 1,000$ (or more), plot the standard deviation in $\hat{p}$ versus p_0 between 0.0 and 1.0. Superimpose points for the Cramer-Rao expectation. Comment: Is Cramer-Rao a good predictor of the standard deviation in this case?
- (g) Now choose $n = 100$ and repeat the study. Comment: What changes do you note in $\langle \hat{p} \rangle$ vs p_0 and $\sigma_{\hat{p}}$ vs p_0 plots? Can you explain the changes?

Submit your assignment by 11:59 PM on Sunday, October 18:

`submit p4730 hw03 hanford_test.py binomial.ipynb`